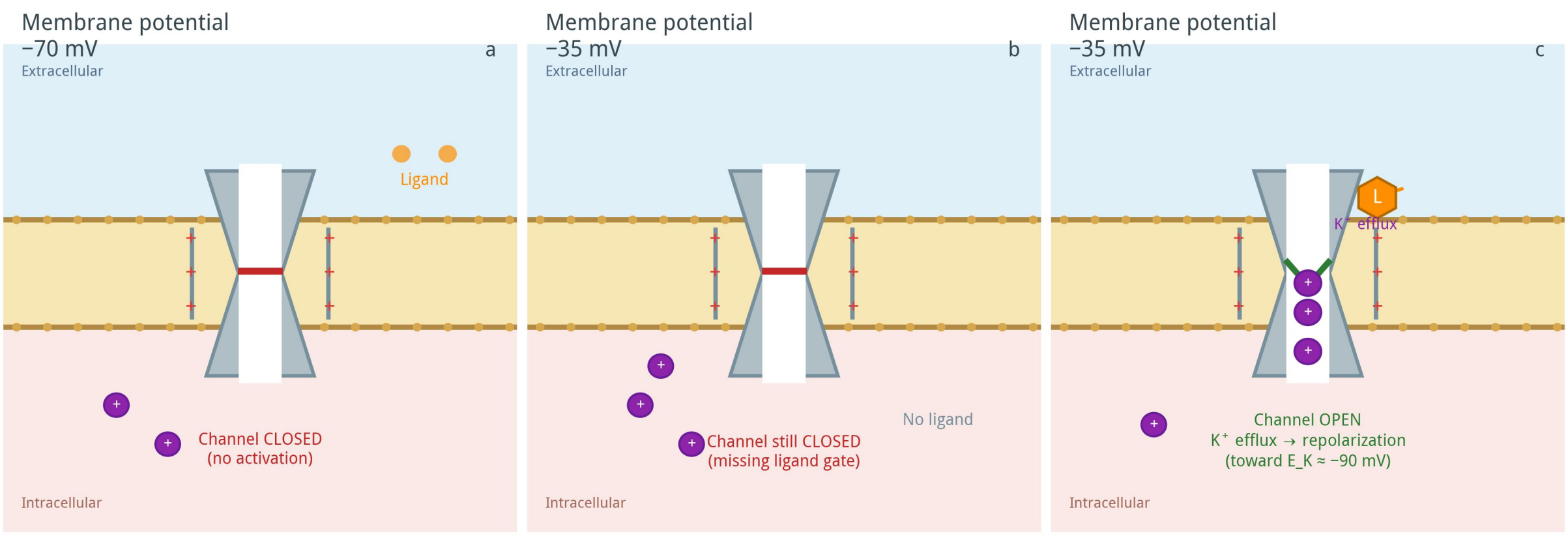

Membrane potential
−70 mV
a
Extracellular
Ligand
Channel CLOSED
(no activation)
Intracellular
Membrane potential
−35 mV
b
Extracellular
No ligand
Channel still CLOSED
(missing ligand gate)
Intracellular
Membrane potential
−35 mV
c
Extracellular
L
K⁺ efflux
Channel OPEN
K⁺ efflux → repolarization
(toward E_K ≈ −90 mV)
Intracellular